Auditing Surface-Order Approaches to GRH: Detailed Proofs and Research Interfaces

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Research draft. GRH is not proved here.

Abstract

We audit proposed surface-order and Hilbert–Pólya arguments for GRH. We retain a conditional spectral-approximation criterion and prove three obstructions: integral hyperbolic matrices have no integer prime-power norms, integral cyclic induction fails already for the Klein four group, and absolute prime tails diverge throughout the proposed critical-strip range. We also exhibit determinant information lost by symmetrization and a direct failure of the supplied diagonal diagnostic. These results identify concrete replacements needed for a complete arithmetic spectral construction. They do not establish GRH. A small Lean 4 companion verifies selected exact arithmetic obstructions; the analytic arguments remain prose proofs.

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1 Scope and conventions

For a primitive Dirichlet character χ of conductor q, write

$$L(s,\chi)=\sum_{n\geq 1}\frac{\chi(n)}{n^s},\qquad \Lambda(s,\chi)=\left(\frac{q}{\pi}\right)^{(s+a)/2}\Gamma\left(\frac{s+a}{2}\right)L(s,\chi),\quad a=\frac{1-\chi(-1)}{2}.$$

The series and Euler product initially apply for $\operatorname{Re} s>1$. Our target is Dirichlet GRH: every zero in $0<\operatorname{Re} s<1$ has real part $1/2$. For $q=1$, use the entire function $\xi(s)=\frac{1}{2}s(s-1)\Lambda(s,1)$ when an entire target is required. Finite-order Hecke characters and general Artin representations require separate constructions; a theorem for one family is not silently a theorem for another.

The supplied surface-order manuscripts [1, 2] propose arithmetic transfer operators and self-adjoint approximants. We isolate valid closure logic and prove obstructions to particular claimed constructions. These obstructions refute intermediate statements, not GRH. The separate `math` repository [3] includes claimed zero-free regions strictly weaker than GRH. Such a region cannot replace the missing critical-line argument.

2 A corrected conditional closure theorem

Theorem 2.1 (Spectral approximation criterion)**.** *Let $U\subset\mathbb{C}$ be connected and open, let f be holomorphic on U and not identically zero, and let*

$$F_N(s)=e^{a_N+b_N s}\det(sI-(\tfrac{1}{2}I+iH_N)),\qquad H_N=H_N^*\in M_{d_N}(\mathbb{C}).$$

If $F_N\to f$ locally uniformly on U, then every zero of f in U has real part $1/2$.

Proof. The eigenvalues $\lambda_{j,N}$ of H_N are real. Thus the zeros of F_N are $1/2+i\lambda_{j,N}$, with algebraic multiplicities; the exponential factor never vanishes. If $f(z)=0$ off that line, choose a closed disk $D\subset U$ about z disjoint from the line, with f nonzero on its boundary. Such a disk exists because the zeros of a nonzero holomorphic function on a connected domain are isolated. Let $m=\min_{\partial D}|f|>0$. Local uniform convergence gives $|F_N-f|<m$ on ∂D for sufficiently large N. Rouché's theorem forces F_N and f to have the same positive number of zeros inside D, contradicting the zero location of F_N. □

Remark 2.2. This is a conditional criterion, not a construction. Self-adjointness alone establishes only the zero location of the approximants. The difficult arithmetic statement is convergence to the particular completed L-function. Pointwise convergence, weak operator convergence, and agreement of finitely many zeros do not supply the displayed hypothesis.

Proposition 2.3 (A usable continuation interface)**.** *Suppose U is a connected domain, the entire functions F_N are locally uniformly bounded on U, and they converge pointwise on a nonempty open subset $V\subset U$ to a holomorphic function f defined on U. Then $F_N\to f$ locally uniformly on U.*

Proof. Montel's theorem makes the family normal. Every subsequence has a further subsequence converging locally uniformly to a holomorphic function g. On V, pointwise convergence gives $g=f$, so the identity theorem gives $g=f$ on U. If the original sequence failed to converge locally uniformly, a compact set, positive error, and violating subsequence would contradict this subsequence property. □

A proposed analytic continuation proof must actually provide the uniform bounds and a genuine overlap V. The disjoint domains $\operatorname{Re} s>1$ and $0<\operatorname{Re} s<1$ are not an overlap.

3 Arithmetic obstructions to the claimed completion

Theorem 3.1 (No integral prime-power geodesic norms)**.** *Let $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ be hyperbolic. Its expanding eigenvalue in absolute value is $u > 1$, and its hyperbolic norm is $N(\gamma) = u^2$. Then $N(\gamma)$ is not an integer greater than one. In particular no closed geodesic arising from such a matrix has length $k \log p$ for a prime p and integer $k \geq 1$.*

Proof. Put $t = |\operatorname{tr} \gamma| \in \mathbb{Z}$, so $t = u + u^{-1}$. If $u^2 = n \in \mathbb{Z}$, $n > 1$, squaring gives

$$t^2 = n + 2 + \frac{1}{n}.$$

The left-hand side and $n + 2$ are integers, so $1/n$ is an integer, contradicting $0 < 1/n < 1$. Since length is $\log N(\gamma)$, length $k \log p$ would imply $N(\gamma) = p^k$, already excluded. □

The statement applies to every subgroup of $\mathrm{SL}_2(\mathbb{Z})$, hence to congruence surfaces. The prime geodesic theorem is an asymptotic counting analogy; it cannot provide the termwise identification claimed in the completion section of [1]. Hecke correspondences of determinant p must not be identified with hyperbolic deck transformations of determinant one.

Theorem 3.2 (Failure of integral cyclic induction)**.** *The trivial character of $G = C_2 \times C_2$ is not an integer linear combination of characters induced from one-dimensional characters of cyclic subgroups of G.*

Proof. Every cyclic subgroup has order one or two. The degree of any such induced character is its index, namely four or two. Evaluation at the identity sends every integer linear combination to an even integer, whereas the trivial character has degree one. □

This distinguishes rational Artin induction from integral Brauer induction. It invalidates the stated cyclic integral reduction, while leaving appropriate forms of these classical induction theorems available for a revised argument.

Theorem 3.3 (Prime-tail obstruction)**.** *For every real $\sigma \leq 1$, the positive series $\sum_p p^{-\sigma}$ diverges. Thus $\sum_{p>N} p^{-\sigma}$ is not a vanishing absolutely convergent tail in the critical strip.*

Proof. It suffices to show $\sum_p 1/p = \infty$, since $p^{-\sigma} \geq p^{-1}$. Suppose the reciprocal-prime series converged. For $0 \leq x \leq 1/2$, $-\log(1 - x) \leq 2x$, for example by integrating $(1 - t)^{-1} \leq 2$ from zero to x. Hence the finite Euler products $\prod_{p \leq X}(1 - 1/p)^{-1}$ would be bounded uniformly in X. Expanding each as a finite product of absolutely convergent geometric series gives the sum of $1/n$ over all positive integers with prime factors at most X. In particular it dominates $\sum_{n \leq \lfloor X \rfloor} 1/n$, which is unbounded. Contradiction. □

Any repair below $\operatorname{Re} s = 1$ must introduce proved cancellation or a regularization, with quantitative control; absolute prime-tail bounds cannot suffice.

4 Determinants and lost arithmetic information

For $T = \operatorname{diag}(\chi(p)p^{-s})_{p \leq N}$ one has exactly

$$\det(I - T) = \prod_{p \leq N} (1 - \chi(p)p^{-s}) = L_N(s, \chi)^{-1},$$

not L_N. On an absolute-convergence region,

$$\log\det(I-T) = -\sum_{k\geq 1}\frac{\operatorname{tr}(T^k)}{k}.$$

The reciprocal orientation matters before any analytic continuation. Moreover Hermitian-part extraction does not preserve this determinant:

$$T = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \frac{T+T^*}{2} = 0, \quad \det(I-T) = 2 \neq 1 = \det(I).$$

An identity for the transfer determinant therefore does not automatically identify the centered characteristic polynomial of a symmetrized operator.

Proposition 4.1 (A direct obstruction to the supplied diagonal diagnostic)**.** *For the $q = 1$ model evaluated at $s = 1/2$, the algorithm that sets $\lambda_p = -\arg(p^{-1/2})$ gives $H_N = 0$ and $F_N(s) = (s-1/2)^{d_N}$. These functions cannot converge locally uniformly to $\Lambda(s,1)$ or $\xi(s)$ on a neighborhood of $1/2$.*

Proof. Every $p^{-1/2}$ is positive, so its argument is zero and $F_N(1/2) = 0$. On the other hand the alternating Dirichlet series $\eta(1/2) = \sum_{n\geq 1}(-1)^{n-1}n^{-1/2}$ is positive: grouping into adjacent pairs gives positive terms and its sum is at least $1 - 1/\sqrt{2} > 0$. The continuation identity $\eta(s) = (1-2^{1-s})\zeta(s)$ gives $\zeta(1/2) < 0$. The gamma factor at $1/4$ is positive and finite, so $\Lambda(1/2,1) \neq 0$, and its pole-removal factor at $1/2$ is nonzero. Even pointwise convergence at that point is impossible. □

5 Consequences for a revised research article

Theorems 3.1, 3.2, and 3.3 require replacements for the associated proof steps, not additional exposition. A credible next theorem would construct operators independently of unknown zeros and establish their exact arithmetic determinant, including ramified, archimedean, and conductor terms. It must then prove locally uniform convergence or the local boundedness in Proposition 2.3. Only then does Theorem 2.1 prove the desired zero location.

Weil/Li positivity, adelic trace methods, and canonical-product approximations are potential alternative interfaces. They move the research obligation to a specific positivity, trace, or approximation theorem; they do not discharge it by analogy. Lagarias [4] and Connes [5] provide relevant primary-source starting points. No novelty or complete proof is claimed for these classical conditional principles.

A Worked geodesic calculations and possible replacements

For trace $t = 3$, the expanding eigenvalue is $(3+\sqrt{5})/2$ and

$$N = \frac{7+3\sqrt{5}}{2}, \qquad \ell = 2\log\frac{3+\sqrt{5}}{2}.$$

This is a legitimate hyperbolic length, but not $\log p$. For $p = 2$, the proposed length $\log 2$ would require $t = 3/\sqrt{2}$, equivalently $2t^2 = 9$, which is impossible even modulo two. The Lean companion verifies this last exact integer obstruction. For arbitrary p^k, the integer argument in Theorem 3.1 is stronger than a numerical search over traces.

A Hecke operator attached to determinant-p double cosets is a different object from a determinant-one deck transformation. A replacement must specify the ambient space, coefficient system, trace class or regularized trace, and the precise arithmetic distribution. Reusing the same symbol for these objects does not construct an identification. An adelic framework may encode prime norms intrinsically; its trace and positivity obligations must then be proved in that framework.

B Character examples and induction bookkeeping

The nonprincipal character modulo three has values $\chi(1) = 1$, $\chi(2) = -1$, $\chi(0) = 0$ and is odd. For scalar weight-zero automorphy under $\Gamma_0(3)$,

$$f(\gamma z) = \chi(d) f(z),$$

the element $-I$ fixes z and has $\chi(-1) = -1$. Hence every such function would satisfy $f(z) = -f(z)$ and vanish. The proposed scalar bundle on the projective quotient is therefore unavailable as written. One can work on a different group or choose a compatible nonzero weight; either change requires rebuilding the operator and trace identity.

Modulo five, the quadratic character has $\chi(2) = -1$. Consequently the one-prime diagonal entry $\chi(2)2^{-1/2}$ is negative. Its 1×1 minor is already negative, so an unrestricted assertion of total nonnegativity of these transfer matrices is false. A positive state constructed as $TT^*/\operatorname{tr}(TT^*)$ discards phase information in this diagonal model: the nonzero character phases disappear from TT^* at real s, although the ramification zero mask remains. Thus positivity of that state cannot by itself recover character-sensitive Euler factors.

For $C_2 \times C_2$, the three order-two subgroups each admit two linear characters. All six induced characters have degree two; induction from the trivial subgroup gives degree four. Negative integer coefficients do not fix the parity obstruction. Rational coefficients can change parity, which explains why rational and integral induction cannot be interchanged. For a monomial representation $\operatorname{Ind}_H^G \theta$, the appropriate Artin formalism yields a Hecke L-function over the fixed field of H; it does not automatically reduce to a product of Dirichlet functions over $\mathbb{Q}$. Every extension must retain the base field and infinity type.

C Local algebra and topology checks

For the upper-Iwahori order

$$\mathcal{O} = \begin{pmatrix} R & R \\ \pi R & R \end{pmatrix}, \quad S = \begin{pmatrix} 0 & \pi \\ 1 & 0 \end{pmatrix},$$

$E_{12} \in \mathcal{O}$, whereas $SE_{12}S^{-1} = \pi^{-1}E_{21} \notin \mathcal{O}$. The displayed shift orientation therefore does not normalize this order. Changing an orientation may repair this isolated calculation, but all subsequent products and gluing maps must be rechecked with that convention.

For $R = \mathbb{C}[[t]]$ and $\lambda \in \mathbb{C}^\times$, there is no unital $\mathbb{C}$-algebra homomorphism sending t to λI in a nonzero matrix algebra: $t - \lambda$ is a unit in R and would be sent to zero. Thus substituting a unit complex phase for a formal uniformizer is not a representation of this allowed complete DVR. An algebra of Laurent polynomials can admit unitary phase evaluation, but it is a different algebra and needs its own construction.

A dessin on a sphere with two vertices and three edges has three faces, by Euler's formula $V - E + F = 2$, not two. Repeating a vertex along a face boundary does not create a new primitive vertex idempotent. In particular, a three-vertex path algebra with arrows between distinct vertices

cannot have the claimed surjection to a six-vertex radical-square-zero cycle algebra: compose with the target's semisimple quotient k^6. Arrows map to zero, since the orthogonal source idempotents force their endpoint supports to be disjoint in the commutative target. The image is generated by at most three orthogonal idempotents, so it cannot be all of k^6. A valid replacement may require a cover or unfolded category rather than a quotient of the stated algebra.

D Why weak convergence and boundary analogies are insufficient

Let P_n be projection onto the nth basis vector of ℓ^2. For every $x \in \ell^2$, $\|P_n x\| = |x_n| \to 0$, so $P_n \to 0$ strongly, but $\|P_n\|_1 = 1$ for every n. Moreover $\det(I - P_n) = 0$ while $\det(I) = 1$. Thus even strong operator convergence does not imply determinant convergence. Trace-norm control is a natural sufficient condition, and must be established for the actual operator family, locally uniformly in s, before invoking Fredholm determinant continuity.

For a Phragmén–Lindelöf argument, one needs uniform control of approximants as the approximation level changes. Polynomial growth of each individual polynomial has constants and degrees depending on N; it does not supply a common bound. Dividing by $(1 + |s|^2)^M$ produces a nonholomorphic expression, to which the maximum modulus principle does not apply. Dividing by $(1 + s^2)^M$ instead introduces poles at $s = \pm i$; a denominator must be holomorphic and nonvanishing on the actual domain. Even after selecting such a denominator, the necessary uniform boundary and horizontal-side estimates are substantive mathematical work.

Likewise, a bound on a density of geodesics is not a bound on a weighted analytic remainder. The full test-function dependence, prime powers, scattering, elliptic and parabolic contributions must be retained. On a cusped finite-area surface the infimum of the injectivity radius is zero. It cannot provide the positive global mesh bound claimed in the source completion argument.

E Spectral approximation can conceal the original hypothesis

Proposition E.1 (Canonical-product converse for the Riemann target)**.** *If RH holds, the entire order-one function ξ admits locally uniform approximations of the form in Theorem 2.1 with diagonal self-adjoint H_N.*

Proof. The Hadamard factorization of ξ, which has no zero at zero, is

$$\xi(s) = e^{a+bs} \prod_{\rho} (1 - s/\rho) e^{s/\rho},$$

with an enumeration for which the canonical product converges locally uniformly. Under RH every $\rho = 1/2 + i\gamma$ has real γ. Each finite product is an exponential of an affine function times a constant times $\prod_{j \le N}(s - \rho_j)$; absorb the nonzero constant into the exponential. Take $H_N = \operatorname{diag}(\gamma_1, \ldots, \gamma_N)$. □

Together with Theorem 2.1, this shows why existence of the approximants, without an independent arithmetic construction, may merely restate RH. It is not progress to import the unknown zero locations into the input spectrum and then prove that the constructed spectrum is real. This proposition is a classical consequence, not a novel proof of RH.

F Research tasks with mathematical acceptance criteria

Track A: repair the determinant interface. Start with the exact diagonal Euler transfer on $\operatorname{Re} s > 1$. Determine which regularized operator determinant could continue the reciprocal Euler product without poles or zeros being silently exchanged. Prove a relation to a self-adjoint operator before replacing the transfer by its Hermitian part. Acceptance requires an identity of functions, not similarity of zero plots. The displayed 2×2 example rejects unrestricted determinant preservation immediately.

Track B: trace norms and normal families. Choose a connected domain containing an Euler-product overlap and the critical strip, after removing the known poles in the target. Supply estimates uniform in N on every compact set. Either construct a trace-class holomorphic family with uniform trace-norm convergence or prove the hypotheses of Proposition 2.3. The divergence in Theorem 3.3 prevents an absolute prime-sum shortcut. Any cancellation estimate must be independently proved without a zero-free region equivalent to the desired conclusion.

Track C: positivity and canonical products. Compare the exact spectral quadratic form with Weil positivity and the Li coefficients for the chosen L-function family. First state the test-function space and regularization, then prove equality of the forms. Positivity of a finite matrix or of finitely many Li coefficients is insufficient; a uniform infinite-family statement is required. Canonical products provide a convergence benchmark, but the preceding converse explains the circularity risk of using zeros as input.

Track D: make formalization match mathematics. The current Lean companion verifies only exact integer and finite-matrix obstructions. It has no definition of Dirichlet L-functions or analytic continuation. A later mathlib layer must use actual complex analytic functions, the standard zeta definitions, local uniform convergence, and self-adjoint operators. An abstract theorem of the form "if an assumed bridge implies GRH, then GRH" is logically valid but would not verify the bridge. Print every theorem's axioms and review its full type and dependencies, not merely its name.

G Computational examples and reproducibility

The Python checks use integers and `fractions.Fraction`, with no random choices, model weights, scientific datasets or floating-point zero finding. They reproduce the Iwahori witness at $\pi = 2$, the determinant change $2 \to 1$, the parity obstruction, and the vanishing diagonal model. The Lean statements verify selected finite calculations and the parity argument in the kernel of a pinned Lean 4 toolchain. Neither executable claims to verify the full complex-analytic arguments in this article.

Both editions share the same core source. The longer edition adds these calculations and research interfaces; extra exposition does not strengthen an unproved hypothesis into a theorem. The source manifest and audit coverage record distinguish full text review, focused review, and unverified dependencies. Publication readiness also requires independent mathematical review, confirmed authorship, and journal-specific preparation.

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